

PIOTROSKI F-SCORE INVESTING

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Introduction

Value investing has been a popular strategy for centuries. An efficient way to do this is by summarizing financial ratios into a qualitative score. Financial ratios are more comparable than absolute numbers, time-invariant, and sector-invariant to some extent, making them perfect candidates to analyze the change in financial health and potential of a company or compare companies with their competitors.

Developed by Stanford accounting professor Joseph Piotroski, the Piotroski F-score is a popular tool to measure the strength of a firm's financial position. The score comprises three categories and nine sub-categories, each worth one point. Higher F-scores represent the best-value stocks. The original author suggested that a company score of eight or nine is considered a good value, and a score between zero and two points is likely a poor value. Researchers found that thresholds from 5-8 also worked, beating their corresponding benchmarks.

Method

We hypothesized that financial robustness and growth would translate to higher stock returns and provide resilience to poor markets. Our independent variable is the F-Score universe filtering, compared to buy and hold of the SPY benchmark over the last three years (7/1/2020 to 7/1/2023). We adjusted portfolio construction and execution models to mimic the market and reduce risk and friction. We study the annual compound return, Sharpe Ratio, Information Ratio, maximum drawdown, and annual standard deviation to examine the strategy's risk-adjusted profitability, robustness, and consistency.

Let's examine how we can implement an F-score strategy with QuantConnect and the MorningStar US Fundamental dataset (<https://www.quantconnect.com/datasets/morning-star-us-fundamentals>).

Universe Selection

We excluded financially "unhealthy" to avoid penny stocks as they are more susceptible to manipulation, making the F-Score or other fundamental-data metrics inaccurate. We selected stocks with a Piotroski F-Score greater than 7 as it balanced financial strength and portfolio diversification. Extreme thresholds filter out most stocks. We did this in the fundamental universe selection model.

```
1. def select_coarse(self, coarse):  
2.     # We only want stocks with fundamental data and price > $1  
3.     filtered = [x.symbol for x in coarse if x.price > 1]  
4.     return filtered  
Click Expand to view more
```

+ Expand

To obtain the Piotroski F-Score, we create the GetPiotroskiFScore method to calculate the Piotroski F-Score of each stock, with its Fine Fundamental object as input argument. It accesses the 9 sub-categories to check if the company meets the following criteria:

1. A positive income
2. A positive operating cash flow
3. Showing increased profitability
4. Income realized as cash flow
5. Low long-term debt
6. Able to repay debts within 1 year.
7. Is fundraising
8. Is profitable
9. Efficiency improvements

```
1. from f_score import *  
2.  
3. def get_piotroski_f_score(self, fine):  
4.     # initial F-Score as 0  
5.     f_score = 0
```

+ Expand

where all the sub-scores are calculated by helper methods stored in f_score.py.

```
1. def get_r_o_a_score(fine):  
2.     '''Get the Profitability - Return of Asset sub-score of Piotroski F-Score  
3.     Arg:  
4.         fine: Fine fundamental object of a stock
```

+ Expand

Portfolio Construction and Execution

We will invest in all the filtered stocks, but we carefully choose the positional sizing and order execution models to reduce risks and transaction costs. We invested the same size in every sector using the SectorWeightingPortfolioConstructionModel. We also used a SpreadExecutionModel to reduce friction costs due to large bid-ask spread in illiquid stocks.

```
1. self.set_portfolio_construction(SectorWeightingPortfolioConstructionModel())
2. self.set_execution(SpreadExecutionModel(0.01))      # maximum 1% spread allowed
```

Reality Modeling

We are unsure about the liquidity of the invested stocks, so we adapt the *VolumeShareSlippageModel* model with price impact on volume invested via a custom security initializer. This could allow the backtest to capture more realistic filling behavior to reflect performance closer to the actual situation.

```
1. # in Initialize
2. self.set_security_initializer(CustomSecurityInitializer(self))
3.
4. class CustomSecurityInitializer(BrokerageModelSecurityInitializer):
    Click Expand to view more
    c
```

+ Expand

Results

Result metrics over the three years are as shown below:

	Piotroski F-Score Investing	SPY Benchmark
Total Return	193.795%	49.592%
Compounded Annual Return	43.224%	14.372%
Sharpe Ratio	1.139	0.722
Information Ratio	1.139	-0.889
Maximum Drawdown	29.900%	26.300%
Annual Standard Deviation	0.293	0.151

Discussion

We can observe that the Sharpe Ratio and Information Ratio of the F-Score strategy are much better than that of the SPY benchmark, implying that each active selection is earning a higher risk-adjusted return. We can also observe that the poor market situation in 2022 had much less impact on the F-score portfolio, which fluctuated less and had a positive trend. This provides qualitative proof of the consistency and robustness of the strategy.

However, the 95% confidence interval of the compounded annual return of the F-Score strategy is (9.197%, 76.637%), while that of the SPY benchmark is (-3.063%, 31.807%). The overlapping 95%CI of the two portfolios showed no significant difference between their annual return. The sample size is too small to draw a high-confidence conclusion that the F-Score strategy earns better than the SPY benchmark. Higher sampling frequency and more extended sampling (backtest) window shall be adapted to draw further conclusions.

The out-performance of the proposed portfolio suggested that the Piotroski F-Score is a great tool to spot under-valued stocks that provide excess returns. The investment strategies using fundamental factor analysis and value investment are consistently working factors while quantifiable to carry out a quantitative analysis. Nonetheless, it is certainly not a silver bullet for various market regimes. It has no prediction, valuation value, or rationale for defending a poor market. Mohr (2012) used F-score and momentum to produce a stable, positive return over the 2008 global financial crisis. Hyde (2018) created a market-neutral strategy by buying high F-score and selling low F-score stocks with equal- and beta-weighting to eliminate systemic risk.

Conclusion

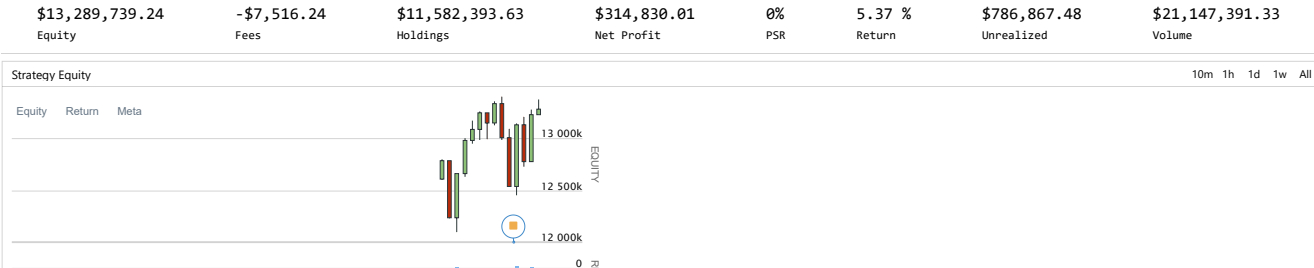
This research has weakly proven the profitability, robustness, and consistency of the Piotroski F-Score filtering.

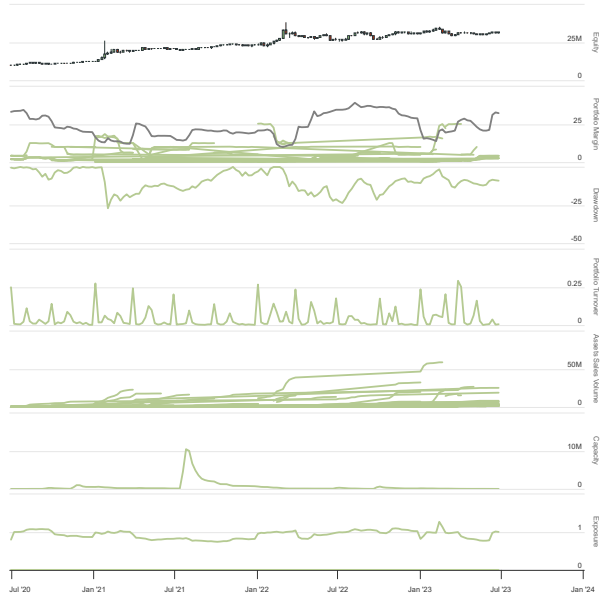
References

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 Hello
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That looks quite demanding in terms of compute power as I am unable to get the same backtest in less than an hour on a free account. Do you have an idea about the recomanded hardware setup to get the backtest done faster?

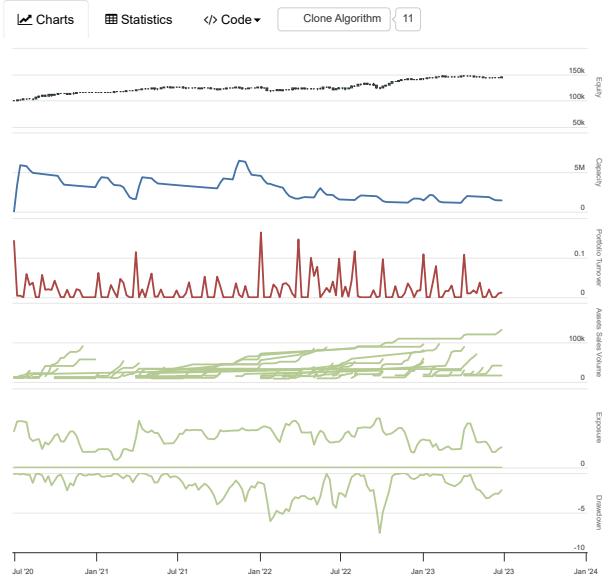
Thanks a lot :)

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https://www.quantconnect.com/u/jared_broad
 Jared Broad
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 Hi AlMoJo, we're working on this to make fundamental backtests 10x faster. It should be ready by mid-September.
 /u/jared_broad
 d)

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